Declaration

Unless otherwise indicated in the text or references, this thesis is entirely the product of my own scholarly work.

Weimar, November 24, 2025

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Pooja Prasad

Abstract

The increasing availability of Generative Artificial Intelligence (AI) systems has transformed how web content is produced, particularly in domains such as reviews and affiliate-driven pages. While these tools enable rapid content creation, they also raise concerns regarding authenticity, manipulation, and declining information quality. This thesis presents a web-scale, temporal analysis of AI-generated and spam content using a dual-model framework that combines classical machine-learning and transformer-based architectures.

A human-annotated dataset developed specifically for this study forms the foundation of the review quality classification. Using this dataset, a BERT-based hierarchical pipeline first identifies whether a webpage contains a review and then classifies each review as either a good or a spam review. In parallel, AI detection is applied across all web pages, covering both review and non-review domains, to distinguish human-written text from AI-generated content using a classical Machine learning model. Together, these components provide a reliable framework for analyzing both authorship and content quality across large-scale web data.

Using the proposed longitudinal dataset of top-ranked pages collected from major search engines over two years, the analysis examines trends in AI usage and spam prevalence on the web. The results show clear shifts over time, including increased AI adoption, variability in spam distribution across search engines, and a weak but measurable correlation between AI-generated text and spam reviews. Moreover, these findings also reveal that although AI-generated and spam content both rise over time, they follow largely independent trajectories. This suggests that modern web manipulation is becoming more diverse not only by automated text generation but also by persistent commercial incentives and evolving optimisation strategies.

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Chapter 1

Introduction

Over the past decade, the web has evolved into a vast and constantly changing information network. Millions of new pages, reviews, and blog posts appear each day, providing instant access to opinions, experiences, and data. However, this rapid expansion has also created new challenges in assessing the credibility of online information. The ease of publishing and the growing use of automated content generation tools have led to an environment in which genuine user feedback and deceptive information often coexist within the same search results [He et al., 2022, Walther et al., 2023].

Search engine optimisation (SEO) also plays a major role in shaping what users see online. Lewandowski et al. [2021] showed that a large fraction of pages found in Google search results are likely to be optimised through SEO techniques such as metadata tuning, keyword targeting, and content structuring. These practices help websites gain visibility but often prioritise ranking performance over informational quality. Combined with the increasing use of automated writing tools, SEO-driven optimisation enables the large-scale creation of content that appears relevant but offers limited substance. As a result, automatically generated and strategically optimised pages can dominate search results, reducing the diversity and trustworthiness of the information users encounter.

A longitudinal study across major search engines demonstrated that the top-ranking results are often occupied by affiliate-driven pages instead of genuine informative sources [Bevendorff et al., 2024c]. Such pages often rely on repetitive templates and commercial linking strategies designed to maximise user clicks. Although affiliate content is not always harmful, its mass production has blurred the line between useful and manipulative information. Prior research indicates that many spam-detection models are evaluated on synthetic or small

datasets and may struggle to generalise to open-web environments [Kaddoura et al., 2022].

Recent shared-task research in Generative AI [Bevendorff et al., 2024a,d] has produced competitive models for distinguishing human and AI-generated text under controlled conditions. However, these systems have so far been evaluated mainly on curated benchmark datasets rather than real-world web data. As highlighted by Reimer et al. [2023], long-term analyses are needed to capture how writing patterns and quality indicators evolve in real search environments.

This thesis addresses that need by examining the overlap between automated text generation and low-credibility online content. It analyses search-engine results comprising 8 million URLs collected across 3 different search engines (Startpage, Bing, DuckDuckGo) over a two-year period (2022–2024) as part of the Webis Affiliate Search Corpus introduced by Bevendorff et al. [2024c]. Using this large-scale dataset, the study explores how textual origin and quality indicators evolve over time within real-world search environments.

A key element of the study is the creation of a human-annotated dataset for evaluating review quality. The annotation process integrates 300 review screenshots, combining users, author and ChatGPT ratings to improve the reliability of the resulting ground truth. This dataset establishes a realistic foundation for assessing text credibility and provides valuable insight into cases where genuine and spam content overlap. Its temporal scope also enables the observation of year-to-year variations in low-quality review material, contributing empirical evidence to ongoing discussions about the trustworthiness of online information. To ensure a consistent labelling, the data were collected and verified through a custom-built web platform where annotators assessed review screenshots based on the visible webpage context. These efforts provide a dependable basis for examining how automated content generation influences the credibility and presentation of online reviews within the spam detection framework.

Building on this corpus, the study addresses two core questions: first, how to determine whether a piece of text was produced by a human or by an AI model, and second, how to evaluate the trustworthiness of user-generated reviews appearing on the web. For AI detection, a classical machine-learning classifier trained on linguistic features distinguishes human-written text from AI-generated content. Next, a BERT-based model capable of capturing deeper semantic and contextual cues is applied for identifying review pages and assessing their quality. Investigating both aspects jointly enables a more complete view of how automation influences the reliability of online information.

Research Goals

This study aims to develop reliable and scalable models for classifying and analysing web text, particularly review pages collected from search engine results. These tasks are important for improving how online content is assessed and ranked. With the growing use of large language models (LLMs), it has become essential to separate human-written text from machine-generated text. Identifying spam reviews is equally important to preserve information quality and user trust on review and e-commerce platforms.

The main research goals to analyse the webpages data for this study are:

RQ1: How do machine learning models perform when applied to distinguish AI-generated and human-written text in large-scale web data?

RQ2: How accurately can transformer-based models identify review pages and classify them as good or spam reviews using human-annotated ground-truth data?

RQ3: What is the measurable correlation between AI-generated content and spam review predictions?

RQ4: How do AI-generated and spam review contents evolve over the two-year period across different search engines?

Chapter 2

Literature Review

Search Engines: Visibility and Quality

Search engines strongly influence the type of content that gains visibility and attention on the Web. Their ranking strategies, transparency levels, and commercial policies vary across platforms. Google, the leading search engine worldwide, relies on advanced algorithms that assess factors such as website authority, topical relevance, and user engagement to determine page ranking [Lewandowski et al., 2021, Reimer et al., 2023]. Despite continuous algorithm updates, studies show that affiliate-linked and low quality pages often occupy top positions in commercial queries, indicating that optimisation strategies continue to enable effective manipulation of search visibility. Analysis across major engines revealed that Google, Bing, and DuckDuckGo all prioritised commercially optimised content over more informative or expert sources [Bevendorff et al., 2024c]. This shows that ranking algorithms reward SEO behaviour such as keyword repetition, backlink density, and structured metadata. Consequently, pages designed to exploit algorithmic preferences gain visibility, while reliable but less optimised material remains buried. This imbalance influences user trust and distorts the information quality that is perceived [Lewandowski et al., 2021, Walther et al., 2023].

Bing applies similar ranking principles but gives greater importance to freshness and multimedia relevance, while DuckDuckGo, though privacy-oriented, displays a comparable level of affiliate and templated content. Long-term observations imply that algorithmic diversity among engines remains limited, with similar ranking mechanisms leading to comparable outcomes [Reimer et al., 2023].

Commercial visibility strongly shapes how content is produced and consumed.

Studies show that online marketing success depends more on ranking position than on the inherent quality of the material [Juhani, 2019]. Businesses therefore invest heavily in SEO campaigns, creating a feedback loop that encourages large-scale production of keyword-heavy or repetitive text. As automation tools improve, such content is increasingly generated through semi-automatic templates or large language models, forming clusters of near-identical pages across domains [Bevendorff et al., 2024c].

From a user perspective, credibility is often evaluated through visual design, tone, and fluency rather than factual verification [Walther et al., 2023]. This cognitive tendency allows well-structured but misleading content to appear trustworthy. As a result, visibility in search results has become an unreliable indicator of quality. The evidence suggests that algorithmic ranking, commercial pressure, and user perception interact to create a web environment where appearance outweighs substance.

Large-Scale Web Analysis and Transparency

While most studies focus on individual engines, large-scale analyses provide a broader understanding of the search systems that evolve. The Archive Query Log (AQL) dataset, introduced by Reimer et al. [2023], contains more than 350 million search queries from over 550 providers collected over twenty-five years. This dataset facilitates works to examine long-term shifts in ranking logic, content bias, and user behaviour. It also promotes open science by offering an independent basis for verifying search transparency, an area traditionally dominated by proprietary data.

The longitudinal work by Bevendorff et al. [2024c] expanded this perspective by analysing SEO spam across different engines and time periods. Their findings demonstrated that affiliate and auto-generated content are now structurally embedded in web search, particularly in product-related domains. Such large-scale results confirm that the degradation of search quality is systemic, not a temporary outcome of isolated algorithmic changes.

Other works have confirmed that optimisation and automation extend beyond search engines to content-sharing platforms. Bevendorff et al. [2024b] found that YouTube's ranking system prioritises promotional or affiliate-linked videos that offer limited informational depth. These cross-platform comparisons suggest that algorithmic incentives, whether in web search or social media, consistently favour visibility-driven content rather than quality-driven content. Together, these findings highlight the importance of datasets like AQL for evaluating the long-term integrity of recommendation systems.

Generative AI Detection

The advent of large language models (LLMs) such as GPT has transformed natural language generation, while transformer-based encoder models like BERT have advanced language understanding [Devlin et al., 2018, OpenAI, 2024]. GPT models produce text that closely mimics human writing, making it difficult to identify synthetic content. Authorship verification, the task of determining whether a text was written by a human or an AI, has therefore become a central research problem.

Recent benchmark efforts such as the "Voight-Kampff" Generative AI Authorship Verification Task [Bevendorff et al., 2024a, 2025] explored this challenge in depth. The official baselines ranged from linear classifiers to zero-shot perplexity methods, while several participant systems used fine-tuned transformer models such as T5 and RoBERTa to achieve higher detection robustness. Results showed high but imperfect accuracy, indicating that AI-generated content can still evade detection when stylistically adapted.

Subsequent work has quantified the real-world scale of this issue. Sun et al. [2025] examined social media data to estimate the prevalence of AI-generated text, finding that it already constitutes a significant share of online communication. The authors emphasised that authorship verification is not merely a technical challenge but a key requirement for maintaining the credibility of digital information. As generative tools become more accessible, the ability to distinguish genuine human writing from automated imitation will remain an ongoing concern.

Spam and Review Detection Studies

Spam detection has evolved from simple rule-based filtering in emails to usage of complex NLP models applied across web platforms. A comprehensive survey by He et al. [2022] summarised key approaches based on linguistic, behavioural, and semantic features. They concluded that while deep learning models achieve strong accuracy, hybrid methods combining structural and contextual cues yield more stable cross-domain performance.

A comprehensive review by Kaddoura et al. [2022] examined existing research on spam content detection and classification. The authors discussed major challenges such as class imbalance, limited interpretability, and the difficulty of applying models across different languages and domains. They observed that several studies have explored hybrid approaches combining hand-crafted linguistic features with deep learning models, which showed better stability

and adaptability across datasets. However, the review also emphasised that more progress is needed in creating balanced datasets and explainable models that can generalise well in real-world conditions.

In a study on social-media spam detection, Sharma and Ramaiya [2022] developed a machine learning framework combining TF–IDF, Part-of-Speech tagging, and Information Gain with classifiers like Support Vector Machines and Artificial Neural Networks. They found that hybrid feature combinations yielded higher accuracy than single-feature approaches. Similarly, Srivastava and Singh [2024] applied linguistic preprocessing and semantic representation to detect spam in multilingual review datasets. These results consistently show that combining linguistic structure with contextual meaning enhances reliability and adaptability.

Integration of Classical and Transformer Models

Transformer-based architectures such as BERT and RoBERTa have become dominant in natural language tasks, but classical algorithms continue to offer strong advantages in efficiency and interpretability [Devlin et al., 2018]. Recently,Sun et al. [2025] showed that traditional models, such as Logistic Regression, Naïve Bayes, and SVM, remain effective for short or structured text, whereas transformer-based models, such as BERT, outperform them on longer or context-rich documents by capturing deeper semantic and syntactic dependencies.

Both He et al. [2022] and Sharma and Ramaiya [2022] showed that combining classical text features with modern deep-learning methods gives more reliable spam detection across different domains. Such designs reduce overfitting and improve decision transparency. They also remain computationally feasible for large-scale deployment, a critical factor in processing millions of documents.

Recent studies show that authorship analysis, spam detection, and search engine optimisation are all related to the same goal — keeping online content real and trustworthy. Bevendorff et al. [2024c] found that SEO-driven spam is often made with semi-automatic tools and has simple, repetitive language with little variety. Such writing is similar to text created by machines. The PAN 2024 authorship verification task [Bevendorff et al., 2024a] also showed that it is difficult to tell human and AI-generated text apart, because machine-written texts can look fluent but often lack meaning and depth.

Reimer et al. [2023] and Lewandowski et al. [2021] explained that search ranking mechanisms unintentionally amplify these effects by rewarding metrics such as keyword density and click rates. Sun et al. [2025] further noted that gen-

erative AI accelerates this process by producing high volumes of coherent but low-value text. Consequently, automated generation has become both a commercial tool and a challenge to online authenticity.

To counter these effects, researchers recommend hybrid detection models that combine linguistic features (e.g., syntax, sentiment, readability) with neural representations learnt from transformer-based models. Studies by He et al. [2022] and Kaddoura et al. [2022] proposed architectures that detect subtle irregularities in style, while Sharma and Ramaiya [2022] and Srivastava and Singh [2024] confirmed that handcrafted features, such as POS ratios and readability measures, still play an important role.

Explainable hybrid frameworks also provide feature-level interpretation through word importance, readability scores, or sentiment analysis. These interpretable features help explain how the model finds manipulative or AI-generated text and support credibility and trust in the results.

Credibility, Ethics, and Information Trust

Recent work shows that people often evaluate information, trustworthiness and credibility based on presentation quality, emotional tone, and writing style rather than factual evidence [Walther et al., 2023]. This tendency is exploited by commercial entities and automated content systems that mimic professional aesthetics to gain legitimacy.

Bevendorff et al. [2024c] observed that content and affiliate networks rely on optimisation techniques such as keyword stuffing, backlink exchanges, and large-scale content templating to prioritise visibility over informational value. While Reimer et al. [2023] emphasised the need for transparency and open-access search data to maintain accountability. In a broader social context, Sun et al. [2025] argued that generative AI challenges existing ethical norms by blurring authorship boundaries and increasing the risk of misinformation. These perspectives converge on the need for combined technical, social, and ethical approaches.

Temporal Patterns and Evolution of Online Content

Temporal analysis, which examines how patterns change over time, provides key insight into the evolution of automation and manipulation. Longitudinal research has shown that affiliate marketing and content automation steadily in-

creases over time across major engines [Bevendorff et al., 2024c]. The repeated use of specific templates, keyword patterns, and low-complexity language suggests that automated optimisation has become a sustained structural feature of the modern web.

Using time-aligned datasets such as the AQL, Reimer et al. [2023] demonstrated that search behaviour, domain frequency, and ranking signals change systematically over time, often following identifiable cycles. This temporal analysis shows that algorithm updates continually reshape visibility patterns and that spammers adapt their tactics in response. Maintaining ongoing temporal monitoring therefore helps to detect new manipulative strategies and evaluate how long countermeasures remain effective.

These findings support the integration of authorship and spam analysis with temporal modelling, enabling early detection of emerging automated trends. By linking linguistic variation with time-dependent behaviour, researchers can track the evolution of authenticity and quality within the digital ecosystem.

Integrating these analytical perspectives provides a broader and more effective framework: authorship verification helps identify the origin of text, spam detection assesses its credibility, and SEO analysis explains its visibility. Temporal and cross-engine research [Reimer et al., 2023] further highlights that manipulation evolves alongside algorithmic updates. Combining these methods therefore enables continuous and adaptive monitoring of web authenticity. Together, these interconnected approaches form the foundation for future systems that ensure transparency, interpretability, and credibility across online content.

Beyond textual analysis, similar automation patterns are also observed in multimodal online environments. AI-generated information now also appears in images, videos, and audio. Due to this, researchers study multimodal information integrity, which means checking the truth and origin of content across different formats. Some works combine language, image, and user-behaviour data to find patterns of false or automated information. These combined methods make detection more reliable and complete [Sun et al., 2025].

On platforms such as YouTube, ranking and recommendation systems work similar to web search engines. They often choose videos that get more clicks and reactions instead of those that give real or useful information [Bevendorff et al., 2024b]. This creates many affiliate networks, where videos and reviews about products look very similar and use the same style or script. These campaigns often reuse the same keywords and messages across websites, videos, and social media posts. They spread across many platforms, hence they are

called cross-platform spam campaigns, and they are harder to detect.

To solve this, new research uses multimodal fusion models that combine language understanding with image or video analysis. Transformer-based systems such as BERT and CLIP can process both text and visual inputs. They help identify mismatches between words and images, for example, when a product description does not match the photo it shows signs of fake or generated content [Devlin et al., 2018].

Some works also use temporal tracking, which means watching the content change over time. This helps researchers understand how fake reviews or AI-made advertisements move from one site to another [Reimer et al., 2023]. By combining time analysis with multimodal features, they can see the spread of content spreads across the web.

However, analysing large amounts of data also raises ethical questions about privacy and fairness. Systems must be transparent and protect user information while checking for false or harmful content [Walther et al., 2023]. Another difficulty is explainability, it is often hard to understand the reasoning behind marking it as spam or good content.

In all, maintaining trustworthy information today requires a multi-layered strategy. Text, images, videos, and user activity together influence which content becomes visible and how it propagates online. Analysing how AI-generated content is shared and reused across different media is important for building reliable systems that can detect and control misinformation across platforms.

Chapter 3

Dataset and Proposed Annotation Framework

This study used a combination of benchmark and large-scale web datasets to train, validate, and apply the classification models. The PAN 2024 Generative Authorship Corpus [Bevendorff et al., 2024a] and a manually annotated subset of the Webis Affiliate-Search Crawl [Bevendorff et al., 2024c] provided the ground-truth data for model training and evaluation. The remaining portion of the Webis Crawl, containing millions of unlabelled web pages, served as the target corpus for large-scale prediction and temporal analysis.

The PAN 2024 dataset contains verified examples of human- and AI-written text, used to calibrate the authorship detection models. The Webis Affiliate-Search Crawl captures search engine results from DuckDuckGo, Bing, and Startpage over a two-year period, offering a realistic representation of online content diversity and search visibility. The following sections describe the datasets, text extraction, feature processing, and annotation procedures used to create consistent and reproducible inputs for classification models.

3.1 Source Corpus: Webis Affiliate-Search Crawl

This study is based on the Webis Affiliate-Search Crawl, a large-scale dataset containing over eight million Search Engine Result Pages (SERPs) collected between 2022 and 2024. The experiments in this research were primarily based on this corpus. It was created to analyse the influence of affiliate marketing and content manipulation on search engine results. Spanning two years, it allows examination of how web content evolved during the rise of large language models (LLMs) and changing SEO strategies.

The data was collected from three search engines — DuckDuckGo, Bing, and Startpage. This allows comparison between platforms in terms of ranking rules, content variety, and repeated pages. The dataset is saved in the standard **Web ARChive (WARC)** format (ISO 28500:2017), which keeps full copies of web pages, including text, HTML code, and metadata. Each crawl folder includes:

- The search query and the HTML page of the search results,
- The top 20 URLs for each query, saved as WARC files,
- Metadata such as date, search engine name, ranking position, and domain information.

This structure enables temporal analysis of changes in search results over time. Each HTML file is a full copy of the web page at the time it was collected, so both the text and page structure can be analysed. In this study, the main focus was on the text content, while metadata such as timestamp and search engine were used for context.

The dataset covers many different websites and topics, giving it wide topical breadth and sufficient temporal coverage to study two connected topics: the rise of AI-generated text in search results and its link with spam or low-quality content.

3.2 PAN 2024 Generative Authorship Corpus

The PAN 2024 Generative Authorship Verification Dataset was used to train and evaluate systems that identify whether a text was written by a human or generated by an AI model. The dataset was created as part of the PAN shared task at CLEF 2024, which focuses on authorship verification and generative text detection.

The PAN 2024 collection contains paired documents, where one is written by a human and the other is generated by an AI model. The texts come mainly from publicly available sources, including news articles, Wikipedia introductions, and short narratives. Each sample in the dataset carries a verified authorship label, ensuring that one text in the pair originates from a real human source while the other is produced by a large language model (LLM). These labels were verified using provenance data and manual checks to ensure their accuracy and reliability.

This dataset has several key strengths:

- **Balanced structure:** Each data pair includes both AI and human text, which helps models learn meaningful linguistic differences without bias toward one class.
- **Domain coverage:** The inclusion of multiple types of writing, from factual news to informal online text, supports broader generalization to different web environments.
- **High-quality validation:** All samples are verified through expert review and metadata tracing, improving the trustworthiness of ground truth labels.

For our study, the PAN dataset served as the main reference for evaluating model accuracy and calibration before large-scale application to web data. Instead of relying on a specific pre-trained model, the benchmark allowed comparison between multiple architectures, including transformer-based and classical approaches, under the same experimental setting.

The purpose of including PAN 2024 was twofold: first, to create a reliable baseline for detecting AI-generated text under controlled conditions, and second, to apply the trained models to real-world search engine data where authorship is unknown. This ensured that performance results from the benchmark could be directly compared to outcomes in real, open-web environments.

Crawl Sampling and Subset Construction

Processing eight million web pages was not practical for spam detection because of limited computing resources, while the complete set was still used for AI authorship analysis. To handle this efficiently, we used a stratified sampling method to build a smaller but balanced dataset. The goal was to keep a wide variety of data across time periods, search engines, and topics while reducing repeated or unnecessary pages.

We treated each crawl as a separate group, and up to 5,000 URLs were randomly selected from each. This process resulted in a stratified subset of about **360,000** unique landing pages covering different combinations of search engine and date. The sample was large enough to study changes across time and between search engines, while keeping the processing time manageable. Before final selection, URLs were cleaned to remove duplicate links so that every retained page represented unique website content and contributed fairly to the overall diversity.

The sample also included a balanced mix of commercial and non-commercial

sites. This helped prevent the spam classification task from focusing too heavily on affiliate websites. Across the three engines, DuckDuckGo and Bing provided close to their full 5,000 URLs per crawl, while Startpage produced fewer results due to shallower crawl limits but still offered useful coverage for comparison.

Data Extraction Pipeline

The extraction of webpage text from WARC files was carried out using a Python pipeline designed for reliable large-scale processing. The system used the `resiliparse` library [Bevendorff et al., 2024c], a toolkit developed by the Webis group that provides fast and fault-tolerant HTML parsing. It follows HTML5 standards and builds a full DOM tree for each page, allowing more accurate text extraction than regular expression methods. The process followed four main steps:

1. **HTML Parsing and Cleaning:** Each file was opened and checked for valid HTML. Only web responses with status codes between 200–299 and the MIME type `text/html` were processed. Non-HTML files such as images, CSS, JavaScript, or PDFs were skipped. The `resiliparse` parser removed layout elements such as advertisements, menus, and headers, keeping only the readable content.

2. **Text Extraction:** The function `extract_plain_text()` from `resiliparse.extract.html2text` was used to extract visible text while ignoring HTML tags. This produced a clean version of the main article or review text.

3. **Metadata Linking:** Each extracted text was linked to its metadata, including the crawl date, search engine, URL, and file path, enabling later analysis of content patterns across engines and over time.

The extraction was parallelised using `joblib` to process millions of pages efficiently. Both the cleaned text and the original HTML were stored together to ensure full reproducibility and allow future reprocessing with updated extraction methods.

Linguistic Feature Extraction

Feature extraction constituted an essential preprocessing step, enabling the transformation of raw inputs into robust feature representations that facilitated

accurate classification. Two different feature sets were used for the two main tasks. For AI versus human authorship detection, lexical features such as Term Frequency (TF) and Term Frequency–Inverse Document Frequency (TF–IDF) were applied to capture word usage and stylistic differences.

For the spam versus good review classification task, a wider range of linguistic and structural features was extracted using the libraries `spaCy`, `textstat`, and `TextBlob` libraries. These included part-of-speech patterns, readability scores, sentiment values, and basic text statistics. These features describe writing style, tone, and grammatical structure, which often differ between authentic and manipulated content. All extracted features were stored in structured `.csv` format along with document metadata. Detailed explanations of each feature type and its role in the classification process are provided in the next chapter 4.

3.3 Developing Custom Annotation Labels and Taxonomy

This section presents the custom annotation framework developed in this study to evaluate spam and review quality. The annotation process was conducted on a manually selected subset of the **Webis Affiliate-Search Crawl** and forms an essential part of the dataset contribution of this research. Three annotation groups were defined to capture complementary aspects of webpage content: review quality classes, non-review website types, and screenshot integrity issues. Together, these categories describe both the textual quality and the functional purpose of each webpage, ensuring consistent and interpretable labeling across diverse content types.

Review Quality Classes: Review quality classes represent different levels of writing style and trustworthiness found in online product reviews. They help separate informative and original reviews from those that are repetitive, copied, or machine generated.

- **Great Review:** Contains detailed product information based on real experience or testing. It often includes original pictures, clear comparisons, and trustworthy references that support the statements made.
- **Decent Review:** Presents organised and clear product descriptions but lacks deeper insight, first-hand testing, or expert judgement.
- **Poor Review:** Offers very limited and repetitive content with little

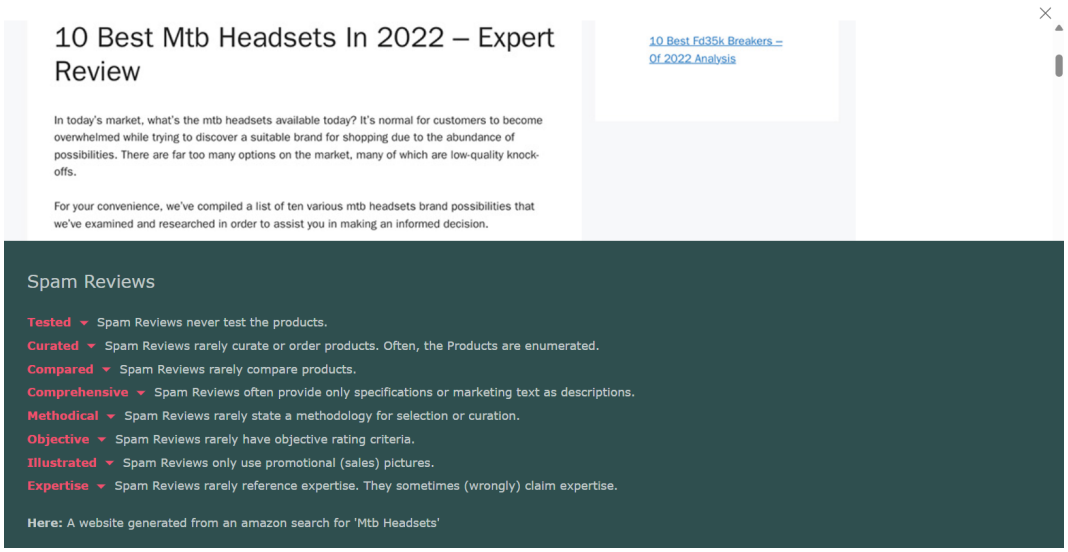


Figure 3.1: Popup instructions for selecting a particular rating in the annotation interface.

useful explanation or evidence. Text is often copied or rewritten from product listings.

- **Spam Review:** Consists of automatically produced or copied text focused on keyword placement rather than real meaning, often created to increase search visibility.

Non-Review Website Types: These categories include webpages that appeared in search results but do not contain direct product reviews. They were included to give a complete view of the types of pages shown by search engines.

- **Webshop:** Online shopping pages where users can buy products or services.
- **Buyer's Guide:** Informative pages explaining product types or giving general advice without reviewing specific items.
- **Other:** Websites that do not belong to the previous groups, such as discussion forums, blogs, or news articles.

Screenshot Integrity Issues: This group includes cases where the webpage content could not be properly read or annotated.

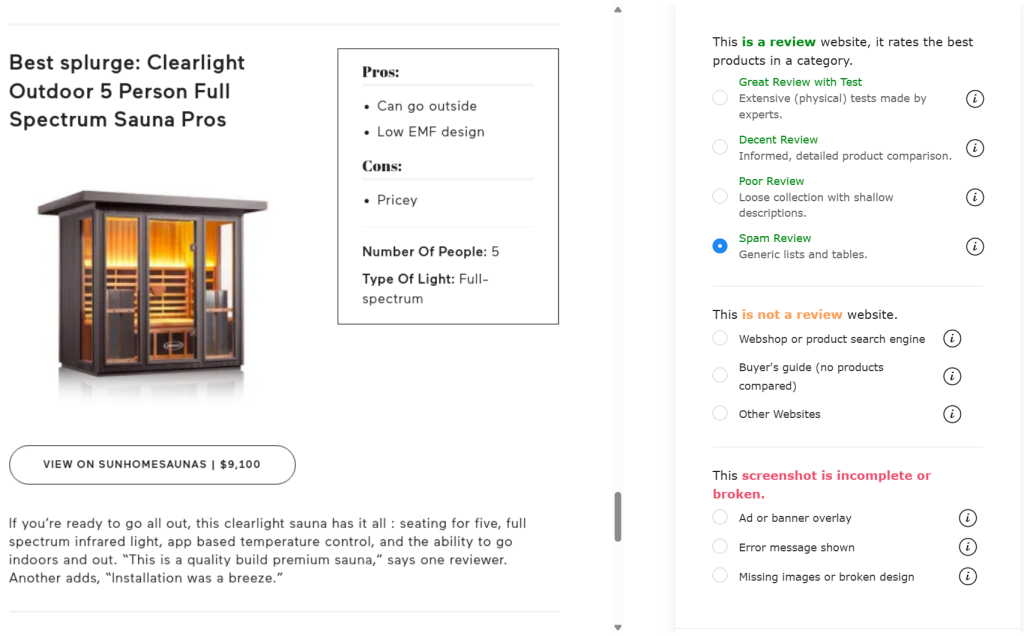


Figure 3.2: Custom web interface developed for manual screenshot annotation.

- **Advertisement Overlay:** The main text is hidden by pop-up ads or banners that block the view.
- **Error Page:** The page shows an error message instead of real content, often due to server or loading issues.
- **Broken Layout:** The webpage appears incomplete because of missing scripts, images, or formatting errors.

Crowdsourced Annotation Protocol

This section details the annotation protocol designed and implemented as part of this study. The procedure aimed to establish a high-quality ground truth for spam and review classification using a manually selected subset of the Webis Affiliate-Search Crawl.

Nine annotators took part in the manual labelling process. A total of 300 webpage screenshots were sampled to represent the diversity of the larger dataset in terms of domain, search engine, and timestamp. The sampling process was stratified: URLs were extracted for each search engine and timestamp pair: Bing, DuckDuckGo, and Startpage across 2022–2024. Within each subset, URLs were randomly selected to include both review and non-review pages. Regular expressions targeting keywords such as review, rating, and comparison

in the URL were used to identify potential review pages. To ensure balance, the final sample included both clear and borderline cases to capture realistic variation in review quality and spam intensity each linked with its corresponding screenshot and metadata.

The 300 webpage screenshots were then divided into three batches of 100, and each batch was reviewed by three different annotators. This setup ensured that every screenshot received three independent labels, while each annotator worked on a single batch to avoid overlap.

Final class labels were decided through majority voting. Screenshots where no agreement was reached were either checked again or removed from the dataset. To measure consistency among annotators, Krippendorff's Alpha was calculated for each batch, showing a high level of agreement.

Annotation Comparison, Verification, and Agreement

To further confirm the reliability of the annotations, a three-way comparison was carried out between the annotators' consensus labels, the author's verification results, and ChatGPT-assisted evaluations [OpenAI, 2024].

The classification of webpage screenshots was performed using a vision–language model based on GPT-5 to assist in large-scale annotation. Each screenshot represented a webpage and was analysed to determine its primary content category. The classification task covered ten possible labels: *Class-1 (Excellent Review)*, *Class-2 (Good Review)*, *Class-3 (Poor Review)*, *Class-4 (Spam Review)*, *Buyer's Guide*, *Webshop*, *Banner or Advertisement*, *Other Website*, *Broken Page*, and *Error or Missing Content* similar to the labels given to users for annotation. To ensure a balanced representation of online material, categories were designed to capture both review-related and general web content types.

The model was guided using progressively refined textual prompts. The initial instruction simply asked the model to "Analyze the screenshot and choose one of the 10 labels given". However, this version often confused similar categories. To improve accuracy, the prompt was expanded to include detailed descriptions and distinguishing features for each label. For instance, the improved instruction defined Buyer's Guide as a comparison article containing phrases such as "Best," "Top 10," or "Buying Tips," while Webshop was identified by elements like product prices, brand names, and "Add to Cart" buttons. Review classes (Class-1 to Class-4) were distinguished based on the clarity, length, and

Table 3.1: Study contribution: Comparison of number of screenshot ratings across annotators, author, and ChatGPT.

Comparison	All Same	Two Same	All Different
All three	59	160	81
ChatGPT vs Author	89	–	211
ChatGPT vs Users	85	–	215
Users vs Author	163	–	137

authenticity of the textual content, ranging from detailed and well-structured user experiences (Class-1) to incomplete or spam-like reviews (Class-4). Additional labels such as Banner or Advertisement, Broken Page, and Error or Missing Content were used for non-review cases and technical anomalies.

During the annotation process, the GPT-5 model evaluated both visual and textual features of each screenshot. It considered layout elements such as number of products shown, table or list structure, banner size and textual cues such as product names, star ratings, opinion phrases, and price tags. For example, a page containing several items with "Top 10 Best" in the heading was classified as a Buyer's Guide, whereas a single product page with detailed feedback and ratings was marked as a Good Review. Each model prediction included both the label and a brief justification to increase transparency. The prompts were iteratively refined between batches to reduce confusion in borderline cases, particularly between medium-quality reviews and mixed promotional pages.

Out of 300 annotated screenshots, 59 showed complete agreement, 160 showed partial agreement, and 81 showed disagreement when all three evaluations were considered. These results reflect the subjective nature of assessing online content quality. Most disagreements occurred in borderline cases, for example when deciding between "decent" and "poor" reviews, or when pages contained both genuine and promotional sections. The ChatGPT-assisted evaluations generally matched well with human labels for clear cases such as strong reviews or obvious spam, but differed for mid-quality or mixed-content pages.

These comparisons showed that review quality assessment is a complex task where human and AI perspectives can complement each other. The final majority-vote labels, reviewed and confirmed by the author, produced a stable and trustworthy ground truth for training and evaluating classification models. A summary of agreement levels is presented in Table 3.1.

Table 3.2: Number of screenshot ratings changed after re-annotation per annotator.

Annotator	Set 1	Set 2	Set 3
Annotator 1	9	10	16
Annotator 2	20	16	20
Annotator 3	19	5	23

Corrections and Re-annotation Statistics

After initial labeling, each annotator re-evaluated some screenshots of their batch. Approximately 20% of the ratings were changed during re-evaluation, primarily where the distinction between buyer's guides and poor reviews was unclear or where banners obscured the core content. Table 3.2 shows the number of changed labels across three rounds.

Agreement Quality via Krippendorff's Alpha

Inter-rater reliability was quantified using Krippendorff's Alpha [Krippendorff, 2004], a statistical coefficient that measures the degree of agreement among multiple annotators while correcting for chance agreement. This metric is widely used in content analysis and annotation studies because it accommodates any number of raters, categorical scales, and missing data. High Alpha values indicate consistent annotation behaviour across raters, suggesting that the labeling scheme was clear and reproducible.

In this study, Krippendorff's Alpha was calculated separately for each annotation batch to assess consistency within and across groups. Sets 1–3 represent the three main annotation batches, while Set 4 corresponds to an initial pilot round used to validate the annotation guidelines and refine category definitions.

- Set 1: **0.80**
- Set 2: **0.85**
- Set 3: **0.89**
- Set 4: **0.81**

Average Krippendorff's Alpha across all sets: 0.84 According to standard interpretation thresholds, values above 0.80 represent strong agreement,

Table 3.3: Comparison of review, non-review, good review, and spam review counts across annotators, Author, and ChatGPT evaluations for the three sets.

Category	Review	Non-Review	Good Review	Spam Review
Set 1				
User 1	58	42	19	39
User 2	50	50	21	29
User 3	47	53	26	21
Author	43	57	24	19
ChatGPT	42	58	36	6
Set 2				
User 1	56	44	27	29
User 2	51	49	27	23
User 3	53	47	32	21
Author	51	49	22	29
ChatGPT	42	58	37	5
Set 3				
User 1	56	44	27	29
User 2	58	42	35	23
User 3	58	42	29	29
Author	50	50	22	28
ChatGPT	52	48	40	12

indicating that the annotators consistently applied the labeling criteria. These results confirm the internal coherence of the annotation protocol and validate the reliability of the final ground-truth labels used for model training and evaluation.

Annotation Results

The final annotation results, summarised in Table 3.3, show the distribution of all screenshots across the four major categories: *review*, *non-review*, *good review*, and *spam review*. These represent the verified and agreed labels after the re-annotation process. Overall, the proportion of reviews and non-reviews remained balanced across all sets, while the distinction between good and spam reviews became clearer following the second evaluation round. The final labels provided a consistent and reliable foundation for subsequent model training and evaluation.

Chapter 4

Experiments

This chapter evaluates the effectiveness of classification models for two tasks: identifying AI-generated text and human-written web pages, and detecting spam within review websites. These experiments use the datasets created in Section 3 and describes each step of the modelling process, including feature extraction, model selection, tuning, and evaluation. The experimental setup was designed to make the models easy to understand, reproducible, and efficient for large-scale use. Special focus is given to combining linguistic features with transformer models so that the system can handle millions of web pages effectively. The chapter is divided into two main sections, one for AI-authorship detection and one for spam review classification.

Feature Engineering

Feature engineering formed an important step in building the classification models. It involved extracting features from the cleaned text that could help the models recognise patterns linked to either AI-generated or spam content. Both linguistic features and statistical text indicators were used, offering a balance between interpretability and accuracy. The following features were used across both the classification tasks:

- **Term Frequency–Inverse Document Frequency (TF–IDF):** TF-IDF quantifies the importance of a word within a document relative to its frequency across the entire dataset [Jones, 1972, Salton and McGill, 1983]. It emphasizes rare but meaningful words. In the AI authorship detection task, TF-IDF effectively identified formal or technical expressions like “delve”, “significance”, that are often seen in AI-generated text. For

spam detection, it highlighted promotional phrases such as "best price" or "top product" commonly seen in spam content.

- **Term Frequency (TF):** Term Frequency counts how often each word or short phrase appears in a document [Luhn, 1957]. It provides a direct representation of word usage within a text without considering the frequency of the same terms in other documents. Unigrams, bigrams, and trigrams were included to capture local phrase patterns and repetition. This feature helps identify repeated or templated wording that often appears in spam content such as "buy now" or "click here", as well as in AI-generated text where phrases are reused for fluency.

- **POS Tag Counts:** Counts of grammatical categories such as nouns, verbs, adjectives, adverbs, pronouns, determiners, and numerals were calculated using spaCy [Honnibal and Montani, 2017]. These act as a linguistic fingerprint that reflects writing style. Spam content often includes many adjectives such as "amazing" or "top" and brand-related nouns, while genuine reviews show a more balanced mix of parts of speech.

- **Normalized POS Ratios:** Raw POS tag counts were divided by the total number of tokens to allow fair comparison between texts of different lengths. This normalization technique originates from early work in stylometry, where grammatical category frequencies were used to identify authors and genres [Argamon et al., 2003, Mosteller and Wallace, 1963]. Spam reviews often show uneven distributions, for example, an unusually high share of adjectives or determiners, and this measure helps to capture such stylistic irregularities.

- **Punctuation Density:** Punctuation Density has long been used as a stylistic marker in authorship and genre studies [Baayen, 2001].This feature measures the number of punctuation marks relative to total word count. Excessive punctuation, especially repeated exclamation or question marks, is common in spam texts that try to attract attention, while authentic reviews tend to follow natural punctuation patterns.

- **Text Length (Token Count):** The total number of tokens in a review indicates the amount of information and writing effort. Spam reviews are often either very short and keyword-heavy or artificially long and repetitive [Bevendorff et al., 2024c]. Measuring document length helped identify such irregular or templated structures.

- **Sentiment Polarity:** Sentiment analysis using TextBlob [Loria, 2018], based on the polarity scale introduced by Pang and Lee [Pang and Lee,

2008], assigns a score between –1 (negative) and +1 (positive). Spam reviews often express extreme emotions, very negative to criticise competitors or very positive to promote products, whereas genuine reviews have more balanced sentiment.

- **Subjectivity Score:** The subjectivity score, also derived from TextBlob [Loria, 2018], estimates the extent to which a review expresses opinion about the subject rather than factual information. Spam reviews often show high subjectivity because they rely on persuasive or emotional language, while good reviews mix factual and personal content more evenly.
- **Flesch Reading Ease:** The Flesch Reading Ease metric [Flesch, 1948] estimates ease of reading a text based on sentence length and syllable count. Spam reviews generally receive very high readability scores because they use short, simple sentences, while genuine reviews include more complex language and details, giving moderate scores.
- **Flesch–Kincaid Grade Level:** The Flesch–Kincaid Grade Level [Kincaid et al., 1975] measures the educational level required to understand a text. Spam reviews typically appear at low grade levels due to short sentences and simple vocabulary, whereas well-written reviews that contain technical terms or detailed reasoning reach higher grade levels.

Modelling Approaches

To evaluate the review classification task, a range of machine learning models were implemented, including both traditional classical classifiers and hybrid transformer-based architectures. Each model was assessed based on performance, interpretability, scalability, and robustness to noisy web content. This section details all models tested during development, including those ultimately not selected for final deployment.

Multinomial Naive Bayes (MNB)

The Multinomial Naive Bayes model, introduced by McCallum and Nigam [1998], is a probabilistic classifier that assumes each feature contributes independently to the class label. It is simple, fast, and often effective for text classification. In this work, MNB was trained using sparse TF and TF-IDF representations of the review corpus. Its low computational cost made it an efficient baseline model. However, its independent assumption limited its accuracy on longer or complex texts. It also struggled with stylised spam reviews that resembled genuine content. Despite these limits, MNB provided a useful reference point for comparing more advanced models.

Support Vector Machine (SVM)

The Support Vector Machine (SVM), introduced by Cortes and Vapnik [1995], is a supervised learning method that separates data by finding an optimal boundary, known as a hyperplane, between classes. A linear SVM uses a straight line (or plane in higher dimensions) as this boundary. It aims to maximise the margin between data points of different classes, which helps the model generalise well to unseen data. In this study, linear SVMs were trained using both TF and TF-IDF feature representations. Platt scaling was applied to convert the raw decision scores into probabilities for easier interpretation. The model achieved strong precision and recall while efficiently handling high-dimensional text data. Among all tested methods, the combination of TF-IDF features with a linear SVM gave the best balance between performance, interpretability, and inference speed. Regularisation using the hyperparameter `C` helped avoid overfitting on noisy web data. The TF-IDF + SVM pipeline was therefore chosen as the final model for AI authorship detection. It consistently captured clear stylistic and lexical differences between human and AI-generated text while remaining computationally efficient.

Logistic Regression (LR)

Logistic Regression, proposed by Cox [1958], is a linear model that predicts class probabilities using a logistic function. It is often used as a simple yet effective baseline in text classification. In this study, Logistic Regression was trained on both TF and TF-IDF vectors. The TF-IDF version performed better by reducing the influence of very common words. While its accuracy was slightly lower than that of the SVM, it provided valuable insight into the most influential words and phrases. Promotional terms such as “best”, “top”, and “review” were linked to spam or AI text, while diverse vocabulary and personal pronouns were more typical of genuine human writing. Logistic Regression confirmed the strength of lexical features but was replaced by the SVM in the final setup due to the latter’s higher robustness and scalability.

Ridge Classifier

The Ridge Classifier, based on ridge regression [Hoerl and Kennard, 1970], was used mainly for spam detection. It was chosen because review quality is an ordinal scale, which means the labels follow a clear order. Instead of giving only a binary output, the model first learns a continuous score that represents how strongly a text belongs to one class, which is then converted into a final label. This approach helps capture small differences between borderline cases, which are common in spam data.

The classifier applies an ℓ_2 regularisation penalty to control large feature weights and improve stability. This property makes it effective for sparse and correlated text features. The Ridge model was trained on TF–IDF vectors derived from annotated reviews. It showed steady performance across validation sets and managed related terms well, such as "buy now" or "discount offer."

However, visual analysis using t-SNE and UMAP indicated that spam and non-spam reviews overlapped considerably, showing that the model found it difficult to form clear class boundaries. Although the Ridge Classifier reached similar F1-scores to Logistic Regression, it did not provide probability outputs, which reduced its interpretability. Still, it remained a strong linear baseline and showed that regularisation improves model stability in large and noisy text datasets.

BERT

The **BERT** model (Bidirectional Encoder Representations from Transformers) [Devlin et al., 2019] is a transformer-based language model that learns the meaning of words from their context on both sides. It reads the entire sentence at once, which helps it understand relationships between words more effectively than traditional one-directional models.

In this work, a version of BERT was used together with additional handcrafted features to improve interpretability and stability on noisy web text. The model combines the contextual output of a pretrained transformer with linguistic and statistical indicators such as TF–IDF weights, part-of-speech ratios, readability scores, and sentiment polarity. These features were concatenated with the transformer embeddings before classification, allowing the system to use both semantic and surface-level information. The architecture includes three main components:

1. A pre-trained transformer encoder (DistilBERT, BERT-base, or RoBERTa-base) that produces a contextual embedding for each document.
2. A feature vector created from TF-IDF weights, POS ratios, readability scores, and sentiment features, selected through mutual information and standardised before training.
3. A classification head that merges both representations, applies dropout, and outputs the final label through a linear layer.

For documents longer than 512 tokens, chunking was applied, and predictions were averaged across chunks.

Models Tested:

- **DistilBERT** [Sanh et al., 2019]: A lightweight version of BERT that retained strong performance while reducing inference time. It was used as the final backbone for large-scale predictions.
- **BERT-base** [Devlin et al., 2019]: The original BERT model trained on English text, providing solid contextual understanding but requiring higher computational resources.
- **RoBERTa-base** [Liu et al., 2019]: An improved BERT model trained on a larger dataset with dynamic masking. It achieved slightly higher recall but increased memory and training time.

Training and Deployment: The pre-trained BERT model was fine-tuned using the HuggingFace `Trainer` API with cross-entropy loss and the AdamW optimizer [Loshchilov and Hutter, 2018]. The top 2,000 features were selected based on mutual information to reduce redundancy.

Deployment: For large-scale spam classification, inference was executed using a SLURM-based batch processing system on the Webis cluster. The final model was applied to 360,000+ documents in parallel jobs using containerized scripts, with predictions stored and aggregated for analysis.

4.1 Experimental Framework for Authorship Classification

Implementation Details

The AI authorship detection experiment used a lightweight and scalable classification pipeline designed for large-scale web inference. The main goal was to identify whether a document was written by a human or generated by a large language model such as GPT-4. This section describes the process used for dataset selection, model training and evaluation.

Dataset and Problem Framing

The PAN 2024 Generative Authorship Verification Corpus served as the benchmark dataset. It contains verified examples of human-written news articles and AI-generated texts from GPT-4-turbo. The task was defined as a binary classification between human and AI text. Before modelling, all documents

were lowercased, punctuation was removed, and only the clean text body was retained.

Train-Validation-Test Split

The dataset was divided into training, validation, and test subsets in an 80–10–10 ratio. Class balance was maintained across all splits to ensure reliable evaluation. The validation set was used to select the best model, and the test set provided a final, unbiased assessment of generalisation.

Feature Representation

To maintain interpretability, the study used statistical text representations instead of deep contextual embeddings. Term Frequency (TF) and Term Frequency–Inverse Document Frequency (TF–IDF) vectors were generated using the CountVectorizer (TF) and TfidfVectorizer (TF-IDF) from scikit-learn. Both unigram and bigram ranges were used, with a feature limit of 10,000 to control dimensionality. These representations preserved lexical and stylistic variation and supported efficient training of linear models.

Model Candidates

Three traditional classifiers were tested across both vectorization schemes:

- **Logistic Regression (LR)** with L2 regularization.
- **Multinomial Naive Bayes (MNB)** with alpha smoothing.
- **Linear Support Vector Machine (SVM)** with Platt scaling for probabilistic output.

This produced six configurations: TF+LR, TF+NB, TF+SVM, TF-IDF+LR, TF-IDF+NB, and TF-IDF+SVM.

Model Selection and Evaluation

Each pipeline was trained on the training set and evaluated on the validation set. For evaluation, metrics such as Accuracy, F1-score were the main evaluation criteria. Among all tested models, the Term Frequency–Inverse Document Frequency and Support Vector Machine combination achieved the highest F1-score and the most stable results across splits. The SVM captured stylistic consistency and subtle lexical patterns characteristic of AI-generated

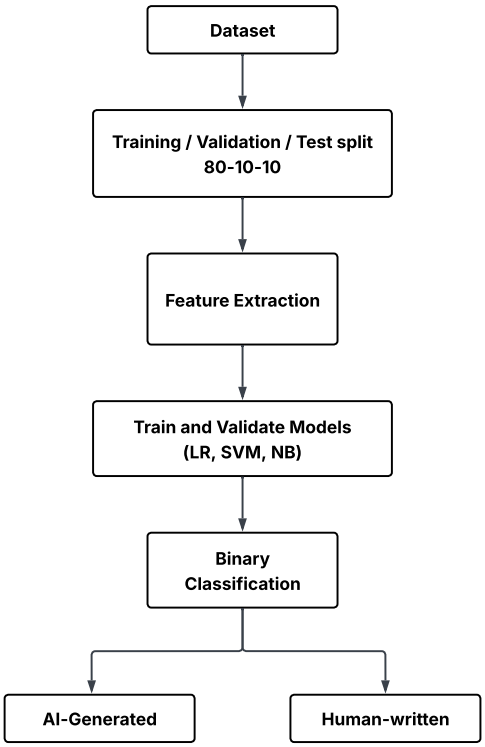


Figure 4.1: Workflow of the AI vs. Human Authorship Detection Pipeline

text, while maintaining transparency and linear scalability.The final model was retrained on the combined training and validation sets (80% + 10%) and tested on the remaining test set (10%).

Model Storage and Reuse

The trained model and vectorizer were stored using the pickle module. This allowed direct reuse during inference without retraining. The prediction script processed each document sequentially from CSV files, applied preprocessing and vectorisation, and produced prediction results with corresponding metadata fields (URL, date, and predicted label).

Inference Pipeline

During inference, the TF–IDF vectorizer transformed the cleaned text into sparse numerical form, which was passed to the trained SVM model for prediction. The system processed data in batches to reduce memory usage. For every input, the output included the predicted label and probability score along

with the URL, file path in WARC directory and timestamp stored in the csv file.

Optimization and Runtime

The inference pipeline was optimised for CPU-based computation using sparse matrix operations and memory mapping. On average, each document including preprocessing, vectorisation, and prediction was processed in about 6 milliseconds. The linear SVM model is efficient on CPU and scales linearly with feature size, hence GPU acceleration was not used. Using GPUs or distributed computation would have introduced extra data transfer overhead without improving speed. The pipeline therefore remained reproducible, energy-efficient, and easy to deploy on standard hardware.

The full system was divided into three independent modules:

- A **training script** for comparing different model–feature pairs.
- An **inference script** for generating predictions on large datasets.
- A **visualisation module** built with matplotlib and seaborn for analysing model confidence, label distribution, and time-based trends.

This modular setup allowed easy debugging, reproducibility, and flexible scaling for future experiments.

Performance Summary

Final outputs from all processed batches were consolidated into a single master file for evaluation and analysis. The TF–IDF and linear Support Vector Machine model demonstrated strong and stable performance on the held-out test data, and it scaled efficiently when applied to the full dataset. Its lightweight design allowed processing on standard CPU hardware without substantial resource requirements.

However, because the model was evaluated only on curated test splits, its behaviour on real-world web data remains uncertain. Differences in writing style, domain, noise levels, and unseen content types may affect generalisation. Although the current framework provides an efficient and interpretable baseline for large-scale authorship detection, additional validation on diverse, real-world samples would be necessary before drawing stronger conclusions about its practical performance.

4.2 Implementation of the Spam Review Detection Model

Overview and Objectives

The spam review detection experiment aimed to identify and classify low-quality or deceptive text within large collections of web documents. The main objective was to separate unreliable reviews from genuine ones and to assess the overall quality of user-generated content that shapes online consumer perception.

A two-stage classification system was developed. The first stage distinguished between review and non-review pages, ensuring that only relevant documents advanced to the next step. The second stage classified these reviews as either Good Reviews or Spam Reviews. This hierarchical setup increased precision by concentrating processing effort on review-related material rather than treating all web pages equally.

The system used a hybrid architecture that combined transformer-based contextual embeddings with handcrafted linguistic and statistical features. Pretrained BERT-family encoders were paired with indicators such as TF–IDF weights, part-of-speech ratios, readability scores, sentiment polarity, and subjectivity measures. This combination allowed the model to use both semantic meaning and surface-level stylistic cues, which helped distinguish authentic writing from manipulative or repetitive content.

Ground-truth annotations were drawn from manually labelled samples collected across search engines and time periods. Each example was tagged according to its clarity, coherence, and alignment with genuine review characteristics. This approach ensured that the model learned from a balanced and diverse dataset covering multiple domains and writing styles.

The following subsections describe the dataset, preprocessing, model training, and large-scale deployment process used for this two-stage framework.

Dataset and Preprocessing

The dataset used for spam detection originated from the Webis Affiliate-Search Crawl, which contains millions of search-engine results collected over two years. A smaller, annotated subset was prepared to create ground-truth labels. These annotations covered different topics, product types, and language patterns. Each record was assigned one of two labels: Review or Non-Review followed by review pages classified into Good Review or Spam Review.

To maintain fair evaluation, the annotated subset was kept separate from the larger prediction set. The labelled data was divided into 80% training, 10% validation, and 10% test portions using stratified sampling to preserve class balance. This ensured stable training and reliable performance assessment. All text samples were preprocessed by lowercasing, normalising punctuation, removing HTML tags, and tokenising with the spaCy NLP library. Texts longer than 512 tokens were divided into segments using a sliding-window approach to fit transformer input limits, while shorter texts were used in full.

Feature Engineering

Feature engineering played a key role in augmenting the semantic representations from transformer embeddings with interpretable linguistic indicators. The following features were extracted and standardized prior to model integration:

- **TF-IDF (1–3 n-grams):** Captures word frequency-weighted importance and detects repetitive keyword stuffing found in spam reviews, such as "buy now", "best price", or "top product".
- **POS Tag Ratios:** Represented grammatical composition through normalised counts of verbs, nouns, and adjectives. Spam text showed higher proportions of adjectives and imperatives.
- **Readability Metrics:** Flesch Reading Ease and Flesch–Kincaid Grade indicated linguistic simplicity or complexity. Promotional or templated text tended to produce extreme scores.
- **Sentiment and Subjectivity:** Sentiment polarity and subjectivity were computed using TextBlob. Spam reviews often showed strong positive or negative tone and high subjectivity.
- **Text Length and Punctuation Density:** Token counts and punctuation ratios described writing rhythm and structure. Spam examples frequently contained short or repetitive sentences with excessive punctuation.

All extracted features were merged into a sparse matrix and reduced to 2,000 dimensions using SelectKBest with mutual information ranking.

Model Selection and Evaluation

A total of eight model configurations were tested to identify the most reliable setup for large-scale spam detection. These included traditional linear classifiers such as Logistic Regression, Support Vector Machine, Ridge Classifier, and Multinomial Naive Bayes, as well as transformer-based models including DistilBERT, BERT-base, RoBERTa-base, and the hybrid BERT architecture.

All models were evaluated using standard performance metrics: accuracy, precision, recall, and macro-F1 score. The linear models served as strong baselines and trained efficiently but performed less effectively on longer or stylistically varied text. Transformer-based models provided deeper contextual understanding but required significantly higher computation time. The hybrid BERT model achieved the best overall performance by combining semantic representations from transformers with interpretable linguistic features. This approach improved generalization and reduced errors on ambiguous or promotional text. Among the tested backbones, the DistilBERT configuration was selected as the final model for deployment because it offered the best balance between accuracy, inference speed, and computational efficiency.

Pipeline Architecture and Workflow

The final detection framework followed a hierarchical design, as shown in Figure 4.2.

Stage 1: Review Identification.: The first binary classifier distinguishes Review and Non-Review content using the BERT model. This filtering step reduced noise and ensured that later computations focused only on review-oriented content.

Stage 2: Spam Classification.: The filtered subset of reviews were then passed to a second binary classifier trained to label each entry as either Good Review or Spam Review. This staged process improved precision and reduced false positives by eliminating irrelevant non-review pages before detailed analysis.

Deployment and Scaling via SLURM and Docker

Distributed inference was required because of the large dataset size. The spam detection pipeline ran on the Webis high-performance computing (HPC) cluster using the SLURM workload manager. Each SLURM job handled a data shard of about 1,000 documents, and processing scaled almost linearly across four compute nodes. The environment for each job was based on a Singu-

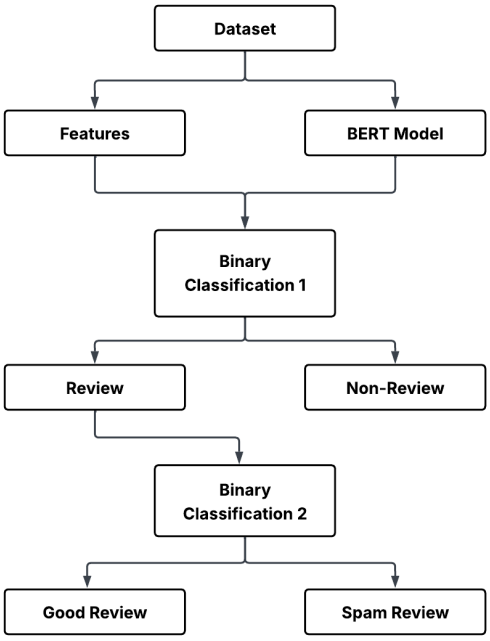


Figure 4.2: Workflow of the Two-Stage Spam Review Detection Pipeline

larity container that was converted from a pre-built Docker image hosted on the private Webis registry-registry.webis.de. This image included all main dependencies including Python 3.12, PyTorch 2.1, Hugging Face Transformers, spaCy 3.7, TextBlob, textstat, and scikit-learn, which ensured consistent and reproducible execution across different nodes.

The master dataset (masterdataset.csv) was automatically divided into equal shards, and each SLURM job processed one shard at a time. Within each job, the system loaded the pretrained binary and multiclass BERT models together with their TF–IDF vectorisers, feature selectors, and scalers serialized using joblib. The text was streamed from disk in small chunks (chunksize = 1000) to reduce memory load, tokenised in mini-batches with DistilBERTTokenizerFast, and passed through the TF–IDF, POS, readability, and sentiment extraction pipelines. Both classification stages were executed one after another, and the predictions were stored as temporary DataFrames before being written to shard-level CSV files. These partial results were automatically merged once all jobs finished.

An adaptive checkpointing system allowed each job to continue from the last

processed record after an interruption. This prevented duplicate computation and saved time during reruns or node reallocation. Model files and intermediate outputs were mounted read-only from the Ceph storage path, which reduced unnecessary I/O operations. The setup also used SLURM's output and error flags to record job logs and error messages, making it easier to monitor progress and recover from failures. Overall, this distributed container-based approach offered a reliable, scalable, and reproducible framework for large-scale spam detection on web data.

Optimization and Runtime Analysis.

The optimized BERT model achieved an average total runtime of about 735 milliseconds per document for both classification stages combined. The first stage (Review vs. Non-Review) required roughly 538 ms, and the second (Good Review vs. Spam Review) averaged 494 ms. The total processing time per document ranged from 356 ms to 1808 ms, corresponding to a throughput of around 80–100 documents per minute per node. In total, about 360,000 web documents were processed across four nodes in parallel. The distributed setup scaled efficiently and maintained stable performance across runs, while the containerised environment guaranteed consistent results and minimized dependency issues.

Conclusion

The spam detection pipeline presented in this study was built using the DistilBERT model, which combines transformer embeddings with interpretable linguistic features. The two-stage setup, first distinguishing review content and then classifying it as good or spam provided a structured approach to analysing large web datasets. When deployed on the Webis HPC cluster with SLURM, the system processed about 360,000 documents using four nodes, achieving a linear scalability and average runtimes of 735 milliseconds per document. Compared to sequential execution, the distributed setup reduced total processing time by roughly 70%.

4.3 Correlation Analysis Between AI-Generated Content and Spam Reviews

The experiments conducted in this section address RQ3, which investigates the measurable correlation between AI-generated content and spam reviews. The relationship between AI-generated content and spam reviews was examined using the Pearson correlation coefficient, which measured the linear association

between AI authorship probability scores and the binary spam labels assigned to each review. The aim was to test whether automated text generation produces linguistic patterns similar to those observed in spam, such as repeated phrases, low information content, or exaggerated sentiment. Three internally generated datasets were combined for this analysis: the AI authorship detection results with per-document probability scores, the spam classification outputs containing both binary and multiclass labels, and the metadata-enriched master dataset. All datasets originated from the models developed in this study and were merged using the common URL field, filtered to include only entries identified as reviews.

A binary variable, Spam Flag was created to indicate spam status, where 1 represented spam or poor reviews and 0 represented all other cases. After aligning timestamps and removing missing entries, the Pearson correlation coefficient between the AI probability score and the spam flag was calculated. The resulting correlation of $r = 0.022$ ($p < 0.001$, $df \approx 2,951,082$) revealed a very weak but statistically significant positive association between the likelihood of AI detection and spam classification. This outcome suggests that, while some overlap exists, the linguistic similarities between AI-generated and spam reviews are marginal. AI-generated content exhibits certain structural and stylistic features occasionally associated with spam, such as repetitive phrasing and limited lexical diversity, yet the correlation strength indicates that the two phenomena are largely distinct. The merged dataset contained about 2.9 million review documents from multiple search engines and time periods, ensuring robust coverage. Month-level grouping was applied to reduce temporal effects related to changing levels of AI or spam content. Despite the weak correlation, the low p-value confirms that the relationship is not a random chance, but the practical effect remains minimal.

Chapter 5

Results and Evaluation

This chapter presents the evaluation of the developed classification pipelines for AI authorship detection and spam review identification. The AI authorship model was applied to more than eight million web documents, and the spam detection system was tested on around 360,000 review-oriented pages. Both systems were assessed using standard evaluation metrics, supported by a correlation analysis that examined links between AI-generated and spam content. Overall, the evaluation of the model demonstrated consistent performance across benchmark datasets and real-world web data.

5.1 Evaluation Strategy and Metrics

The evaluation aimed to measure the accuracy and consistency of each model in distinguishing between the target classes in binary classification tasks. Since real-world web data is often imbalanced and noisy, several complementary metrics were used to ensure a fair and reliable assessment.

The main evaluation measures were Accuracy, Precision, Recall, and F1-score. Each metric provides a different view of model performance and helps understand specific strengths and weaknesses.

Precision: Precision measures the proportion of correctly predicted positive instances among all instances predicted as positive. It shows the model's ability to avoid false positives:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP represents the number of true positives and FP the number of false positives.

Recall: Recall measures the proportion of actual positive instances that were correctly identified by the model. It reflects the ability to avoid false negatives:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN denotes the number of false negatives.

F1-score: The F1-score combines Precision and Recall into a single value by taking their harmonic mean. It is especially useful when the dataset is imbalanced or when both types of errors are important:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Together, these metrics evaluate the model performance effectively. They measure the overall accuracy, class-specific sensitivity, and the trade-off between precision and recall. Using these set of measures ensured that the evaluation considered not only predictive accuracy but also robustness, interpretability, and reliability qualities that are essential for deploying models at web scale.

5.2 Performance Evaluation of Authorship Detection Models

Experimental Setup and Methodology

To address the first research question (RQ1), a set of classic machine learning classifiers was implemented and evaluated for the binary task of authorship attribution. The goal was to distinguish between text segments written by humans and those generated by Large Language Models (LLMs). The PAN 2024 Generative Authorship Verification Corpus served as the benchmark dataset. It contains an equal number of human-authored and AI-generated news-style documents, ensuring a balanced setup for model training and evaluation.

Before modelling, the text data were normalised and converted into numerical representations using Term Frequency (TF) and Term Frequency–Inverse Document Frequency (TF–IDF) features. Both unigrams and bigrams ($n = 1, 2$) were extracted, and feature selection was limited to the 10,000 most informative terms.

Three classifiers were tested: Logistic Regression, Multinomial Naïve Bayes, and Support Vector Machine (SVM) with a linear kernel. The PAN dataset was split into training (80%), validation (10%), and test (10%) subsets using

Table 5.1: Comparison of the model–feature combinations for AI detection. (TF = Term Frequency, TF–IDF = Term Frequency–Inverse Document Frequency)

Model Configuration	**Accuracy**	**F1**
Support Vector Machine + TF–IDF	**0.997**	**0.997**
Logistic Regression + TF	0.993	0.993
Logistic Regression + TF–IDF	0.990	0.990
Multinomial Naive Bayes + TF	0.990	0.990
Multinomial Naive Bayes + TF–IDF	0.988	0.988
Support Vector Machine + TF	0.986	0.986
Best Performing Model	**Support Vector Machine + TF–IDF**	

stratified sampling to maintain class balance. Model selection was guided by performance on the validation set, and the final metrics were reported on the unseen test data.

Comparative Evaluation and Model Selection

Models trained with TF–IDF features consistently outperformed raw term frequencies. The best results were obtained with the linear Support Vector Machine (SVM) combined with TF–IDF representations. This configuration achieved the highest accuracy and F1 scores across validation and test sets. The linear SVM works well because it can draw clear and stable boundaries in TF–IDF feature space, allowing it to distinguish human-written and AI-generated text even when the differences are subtle.

Evaluation on the Test Set

The selected model was evaluated on the reserved 10% test partition (109 human, 109 AI). It achieved:

- **Accuracy:** 0.995
- **Macro F1:** 0.995
- **Macro Recall:** 0.995
- **Macro Precision:** 0.995

$$\text{Confusion Matrix: } \begin{bmatrix} 109 & 0 \\ 1 & 108 \end{bmatrix}$$

Table 5.2: Test Set Evaluation for SVM + TF–IDF Model (80/10/10 Split)

Metric	**AI Class (0)**	**Human Class (1)**
Precision	0.99	1.00
Recall	1.00	0.99
F1	1.00	1.00
Average Accuracy (Macro)	**0.9954**	

The model achieved an overall accuracy of 99.5%, with only a single misclassification where a human-authored document was mistakenly identified as AI-generated.

In summary, the linear Support Vector Machine (SVM) model showed clear separation on the benchmark test set between human and AI-generated text, with balanced precision and recall for both classes. Its transparent structure and efficient computation make it suitable for large-scale use.

Key Findings from AI vs Human Attribution Trends

Before analysing temporal patterns, the dataset underwent several preprocessing steps to ensure consistency and reliability. First, all documents containing valid text content and metadata were selected from the Webis Affiliate Search Corpus. Duplicate URLs appearing in the same month were removed to avoid counting the same page more than once. This month-level grouping provides the foundation for all trend visualisations and correlation analyses presented in this section.

The first part of RQ4, which examines the temporal evolution of AI-generated content across search engines, is addressed in this section. The analysis of temporal patterns focused on changes in the proportion of AI-generated and human-written pages over time. In the plot below, the x-axis represents months, and the left y-axis (dual scale) shows the number of URLs classified as human-authored (blue bars) and AI-generated (red bars). These values correspond to the relative share of each content type in the search index during each period. The right y-axis presents the AI Attribution Rate (AAR) as a percentage, drawn as a black line with circular markers.

The AI Attribution Rate (AAR) is a normalised measure of AI-generated content and is defined as:

$$\text{AAR} = \frac{n_{\text{AI}}}{n_{\text{AI}} + n_{\text{Human}}} \times 100$$

where n_{AI} and n_{Human} denote the number of AI-predicted and human-predicted URLs, respectively, for a given month. The AAR metric enables month-to-month comparison independent of total URL volume fluctuations.

This approach follows the definition and methodology outlined in recent web-scale authorship attribution research on AI-generated content detection [Sun et al., 2025]. Monitoring AAR over time enables observation of how the share of AI-authored content changes in relation to human-generated material within search engine results. The Pearson correlation coefficient (r) and its corresponding p quantify both the strength of the temporal trend in AAR and whether the observed change is statistically significant.

DuckDuckGo

The analysis of DuckDuckGo results shows a clear but gradual rise in AI-generated content over time. In February 2023, the search index was almost entirely dominated by human-authored pages, with around 230,000 human-written URLs and only about 200 AI-labeled ones, giving an AI Attribution Rate (AAR) of roughly 0.25%. From mid to late 2023, the number of AI-generated URLs increased steadily from around 170 to nearly 610, while human-written pages declined to about 130,000. Around October to November 2023, the proportion of AI-generated content began to grow more noticeably, marking the start of a slow shift toward generative material. The highest values appeared in January 2024, with approximately 700 AI-generated URLs compared to 145,000 human-authored ones, resulting in an AAR peak of about 0.5%. Although the share of AI-generated content increased, DuckDuckGo's overall AAR stayed below 1% throughout the study period. This indicates that human-written pages continued to dominate the index, even as generative content became more common. The pattern observed in DuckDuckGo closely followed the trend seen in Bing (Figure 5.2), indicating a consistent rise in AI-generated text across different search engines, with differences mainly in scale rather than direction. The correlation analysis yielded a Pearson coefficient of $r = 0.940$ with $p = 0$, confirming a strong and statistically significant positive relationship between time and the AI Attribution Rate (AAR). This strong correlation suggests that the proportion of AI-generated content on DuckDuckGo increased steadily over time, even though the absolute volume of AI-labeled pages remained comparatively low.

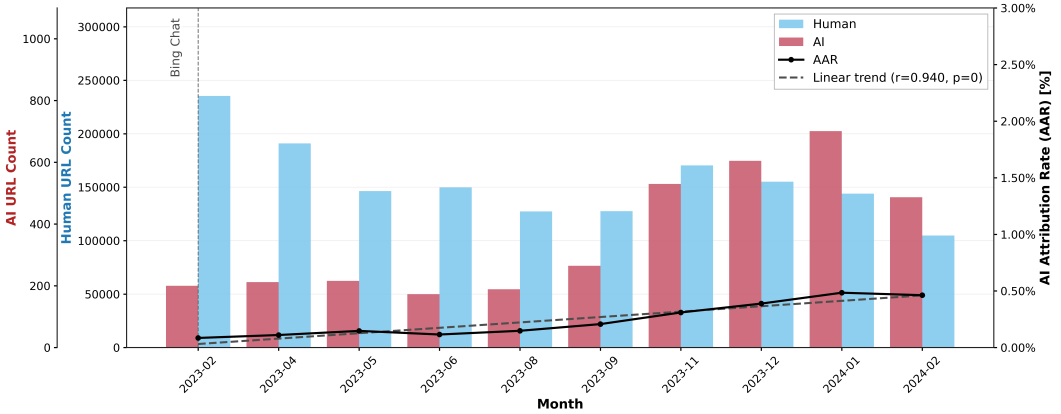


Figure 5.1: Temporal evolution of AI and human content over two years on DuckDuckGo with AI Attribution Rate (AAR)

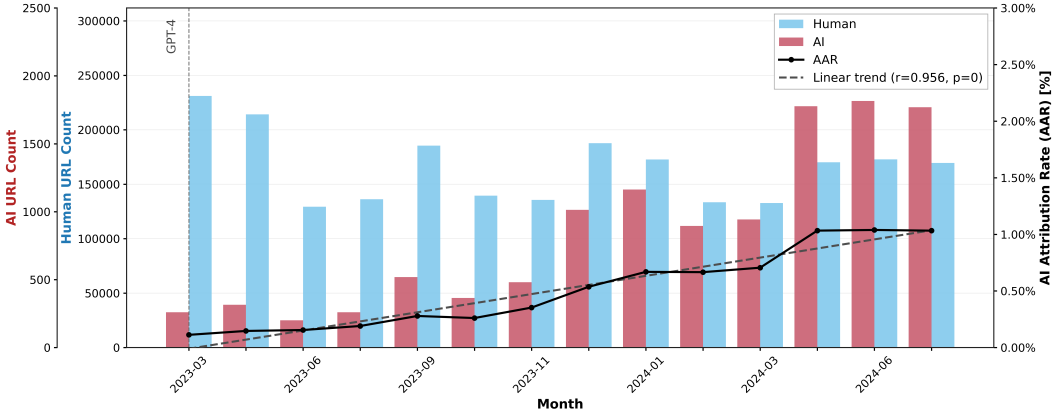


Figure 5.2: Temporal evolution of AI and human content over two years on Bing with AI Attribution Rate (AAR)

Bing

The results for Bing show stronger monthly variations and a more visible rise in AI-generated content compared to DuckDuckGo. In early 2023, Bing indexed about 230,000 human-written URLs and roughly 250 AI-labeled ones, giving an initial AI Attribution Rate (AAR) of around 0.2–0.3%. During mid-2023, both human and AI URL counts dropped, with around 130,000 human and 230 AI entries, keeping the AAR below 0.5%. From late 2023 onwards, a sharp increase in AI-generated content was observed, the number of AI-labeled URLs rose from around 500 in November to more than 1,200 by early 2024, pushing the AAR to about 0.7%, which then stayed stable until March.

Between March and July 2024, AI URLs reached their highest level, between

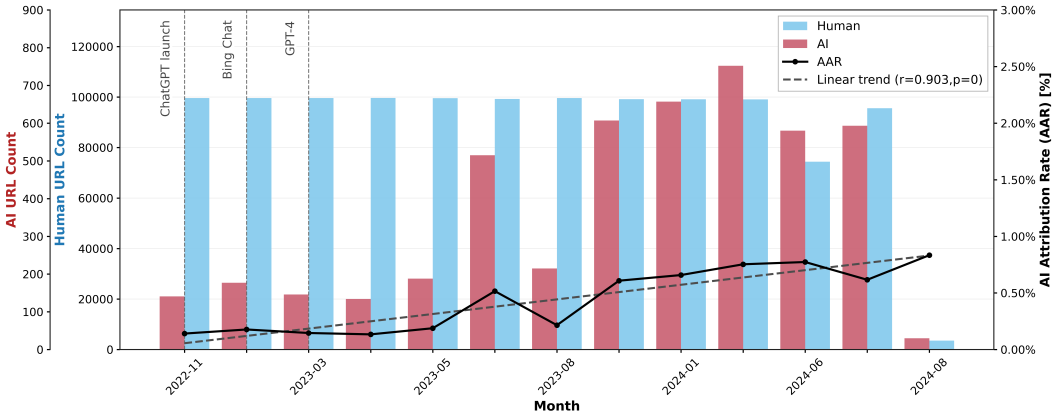


Figure 5.3: Temporal evolution of AI and human content over two years on Startpage with AI Attribution Rate (AAR)

1,750 and 1,850, and the AAR peaked around 1.0–1.1%, representing roughly a doubling compared to the previous year. Overall, the findings indicate a steady and clear growth of AI-generated content within Bing's index, although it still accounted for only a small portion around 1% of the total number of indexed pages. The correlation analysis yielded a Pearson coefficient of $r = 0.956$ with $p = 0$, indicating a very strong and statistically significant positive relationship between time and the AI Attribution Rate (AAR). Despite moderate fluctuations in individual months, the consistently upward trajectory confirms a clear long-term increase in the proportion of AI-labeled content indexed by Bing.

Startpage

The analysis of Startpage data shows a slower but steady rise in AI-generated content, accompanied by minor fluctuations over time. In the early phase between late 2022 and early 2023, the index was dominated by human-authored pages, averaging around 100,000 URLs per month, while AI-labeled pages remained below 170, resulting in an AI Attribution Rate (AAR) under 0.3%. From mid-2023 onwards, AI-generated content began to increase, reaching nearly 510 URLs by June 2023, while the number of human-authored pages stayed constant at about 100,000. During this period, the AAR briefly exceeded 0.5%, marking the first notable phase of growth. Between October 2023 and March 2024, both AI and human URL counts remained stable, with AI-generated pages consistently between 600 and 700 per month, keeping the AAR close to 0.8%. A temporary decline followed in mid to late 2024, when AI-labeled URLs dropped to below 300, leading to a slight decrease in AAR to around 0.7–0.8%. Overall, AI-generated content on Startpage increased gradually but remained under 1% of total indexed URLs, showing that while

generative text is becoming more common, human-authored pages continue to form the majority of the search index. The correlation analysis yielded a Pearson coefficient of $r = 0.903$ with $p = 0$, indicating a very strong and statistically significant positive relationship between time and AI Attribution Rate. This result confirms a clear upward linear trend in AI-labeled content on Startpage, reflecting a consistent and measurable increase in the share of AI-associated pages over the examined period.

Comparison of Trends Among Search Engines

The comparison across all three search engines revealed a gradual but consistent rise in AI-generated content within web search indices. Overall, the upward trend in the AI Attribution Rate (AAR) confirms that generative text is becoming a regular part of online content. However, the scale and timing of this increase differ across engines. Bing showed the earliest and most noticeable growth, reaching an AAR of around 1% by mid-2024. DuckDuckGo maintained the lowest levels, with AI-generated material staying below 1% throughout the period and increasing only modestly. Startpage showed a delayed but steady rise beginning in mid-2023, peaking near 0.8–0.9% before experiencing a slight decline. These variations reflect engine-specific factors such as index size, crawling policies, and user demographics, which influence how quickly AI-generated content becomes visible in each system. Altogether, the results suggest that while human-authored text continued to dominate search results, AI-generated content is steadily gaining presence, pointing to an ongoing shift in the composition of web information.

Distribution of AI-Generated and Human-Written Content Across Search Engines

To provide an overview of the large-scale inference results, Table 5.3 summarizes the total number of analysed URLs per search engine along with the proportion of AI-generated and human-authored pages. Predictions were obtained from the fine-tuned SVM + TF-IDF classifier applied to the subset of the Webis Affiliate-Search Crawl. Across all search engines, the proportion of AI-generated content remained below 0.5%. The combined Bing and DuckDuckGo dataset contained around 21,892 AI-labeled pages (0.33%), while Startpage contributed approximately 5,007 pages (0.36%). Although these proportions are small, they demonstrate that AI-generated text was consistently present across all engines. The concentration of such pages within specific time periods and domains provided the basis for the subsequent temporal trend analysis.

Table 5.3: Distribution of AI-Generated and Human-Written Content Across Search Engines

Search Engine	Total URLs	AI-Generated	Human	AI Share [%]
Bing+DDG	6,698,686	21,892	6,676,794	**0.33**
Startpage	1,374,681	5,007	1,369,674	**0.36**
Total	8,073,367	26,899	8,046,468	**0.34**

Conclusion

The linear Support Vector Machine using TF–IDF features achieved strong, balanced performance on the held-out test set for distinguishing AI-generated from human-authored text. Its efficiency, interpretability, and robustness make it suitable for both controlled benchmark testing and large-scale real-world deployment. The temporal findings indicate a steady growth in generative content, underscoring the need for adaptive detection, attribution, and moderation methods within search and content management systems. Overall, the results highlight the practicality and importance of interpretable machine learning approaches for precise monitoring of AI-generated material in the evolving landscape of large language models.

5.3 Evaluation of Spam Detection

The spam detection pipeline directly addresses RQ2, which investigates how effectively transformer-based models can identify review pages and classify them as good or spam reviews using human-annotated ground-truth data. To achieve this, a structured two-stage prediction framework was designed to separate general web content from user-generated reviews and to further assess the credibility of those reviews.

In Stage 1 (Review vs Non-Review), the first classifier separated genuine user reviews from non-review web text such as blogs, news articles or metadata fragments. This ensured that only user-generated review content advanced to the next stage for quality assessment. The classifier used DistilBERT embeddings together with lexical and linguistic features, including term frequency–inverse document frequency (TF-IDF), part-of-speech ratios, and readability scores.

In Stage 2 (Good Review vs Spam Review), the second classifier identified whether the review was legitimate or spam. Spam reviews were defined as those showing promotional intent, repeated template phrases, excessive punctuation,

or overly positive sentiment. Genuine reviews were characterised by topic relevance, coherence, and balanced language.

This two-stage prediction process is capable of distinguishing four main categories: review, non-review content, genuine user reviews, and spam reviews. This structure also supported transparent model evaluation, each stage was tested separately to assess classification precision, recall, and overall reliability. The two-stage process ensured that review identification and spam detection were both systematically addressed, forming the foundation for analysing review quality and credibility in large-scale web data.

Chosen Model and Feature Set

Based on the comparative evaluation, the DistilBERT configuration was chosen as the final model for both binary stages because it provided a strong balance between accuracy, interpretability, and computational efficiency. DistilBERT retained most of BERT's contextual understanding while offering faster inference, which made it suitable for large-scale processing on the Webis cluster.

To complement the contextual embeddings with interpretable linguistic information, several handcrafted features were added:

- **TF-IDF vectors (1–3 n-grams):** captured keyword emphasis and repetitive phrases typical of spam or templated reviews.
- **Part-of-speech ratios:** quantified stylistic features such as overuse of adjectives and imperative verbs common in promotional text.
- **Sentiment polarity and subjectivity:** measured emotional intensity and bias, helping to identify manipulative or exaggerated opinions.
- **Readability metrics (Flesch Reading Ease and Flesch–Kincaid Grade):** detected overly simple or verbose sentence structures typical of spam.

This hybrid approach combined deep semantic representations from the transformer with surface-level linguistic cues, improving the model's ability to detect deceptive, repetitive, or low-quality content across different web domains.

Detailed Evaluation of DistilBERT

The final DistilBERT model was evaluated through a two-stage classification process. The first stage performed a binary classification to distinguish between Review and Other content, ensuring that only relevant documents were

Table 5.4: Comparison of Hybrid Transformer Models for Spam Detection (Full Evaluation Set)

Model	**Task**	**Accuracy**
BERT	Binary	0.74
	Binary (Good vs. Spam)	0.55
DistilBERT	Binary	**0.78**
	Binary (Good vs. Spam)	**0.63**
RoBERTa	Binary	0.72
	Binary (Good vs. Spam)	0.59
Best Performing Model	**DistilBERT**	

retained for detailed analysis. The second stage further classified the identified reviews into Good Reviews and Spam Reviews, assessing the overall quality and authenticity of user-generated content. The class distribution and corresponding evaluation metrics for both stages are summarised below.

The dataset showed moderate class imbalance, with a slightly higher number of review-related documents compared to non-review text and a larger share of Good Reviews than Spam Reviews. The following sections describes the evaluation results for each stage in detail.

The binary classification stage reached an overall accuracy of 78%, with 58% review pages and 42% non-review pages. This indicates that the model effectively distinguishes review pages from general web content, maintaining stable performance without strong bias toward either category.

In the second stage, which focused on review quality classification, the model achieved 63% accuracy with 33% Good Reviews and 25% Spam Reviews. It showed higher recall (0.73) for Spam Reviews than for Good Reviews (0.53), which means it successfully detected most spam instances while allowing a few genuine reviews to be misclassified.

Interpretation. The confusion matrices further confirm that the first classifier produces balanced predictions, while the second one slightly over predicts the Spam Review label. This cautious bias helps identify potentially misleading or repetitive reviews more reliably. Overall, the DistilBERT model works well for identifying review pages but shows weaker performance in distinguishing good from spam reviews. While it remains computationally efficient for large-scale analysis, the second stage highlights the need for more specialised or fine-tuned models.

Table 5.5: Class Distribution for Two Binary Tasks

Task	Class	Proportion (%)
Stage 1: Review Detection	Review	58.00
	Other	42.00
Stage 2: Review Quality	Good Review	33.00
	Spam Review	25.00

Table 5.6: Performance Comparison of the Two Binary Classification Stages

Metric	Stage 1: Review vs Non-Review	Stage 2: Good vs Spam
Precision	0.78 (Review)	0.61 (Spam Review)
Recall	0.81 (Review)	0.73 (Spam Review)
F1	0.80 (Review)	0.66 (Spam Review)
Precision	0.77 (Non-Review)	0.66 (Good Review)
Recall	0.75 (Non-Review)	0.53 (Good Review)
F1	0.76 (Non-Review)	0.59 (Good Review)
Accuracy	**0.78**	**0.63**

Review vs Non-Review $\begin{bmatrix} 21 & 7 \\ 6 & 26 \end{bmatrix}$ Good vs Spam Review $\begin{bmatrix} 8 & 7 \\ 4 & 11 \end{bmatrix}$

Figure 5.4: Confusion matrices for both classification stages.

Temporal Evolution of Spam (2022–2024)

The second part of RQ4, which examines the temporal evolution of spam content across search engines, is addressed in this section. Spam prevalence was monitored over a two-year window (August 2022 – June 2024) across three search engines: DuckDuckGo, Bing, and Startpage. The temporal evolution was complemented by a correlation analysis between the spam rate and time index, using the Pearson correlation coefficient r. Monthly spam rate was computed as:

$$\text{Spam Rate} = \frac{n_{\text{Spam Reviews}}}{n_{\text{Total Reviews}}} \times 100.$$

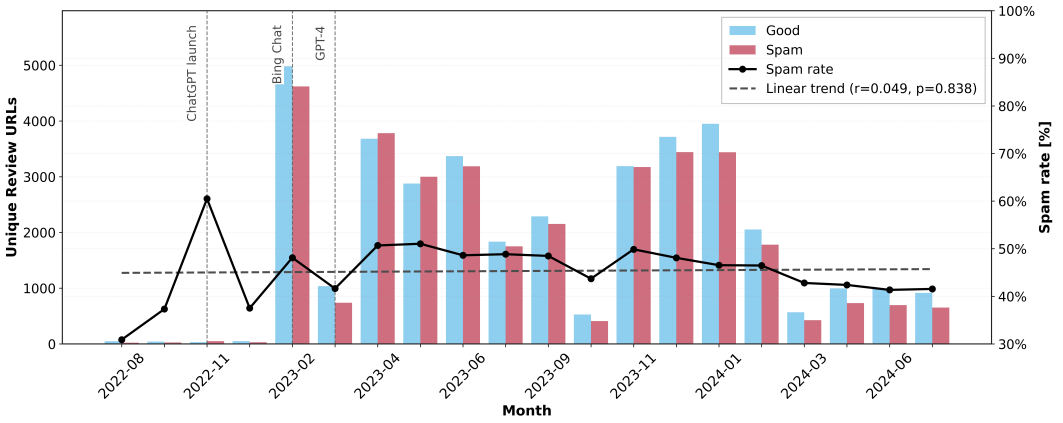


Figure 5.5: Temporal evolution of Spam vs. Good Reviews on DuckDuckGo with Spam Rate (%)

Key Findings

Spam activity increased notably between late 2022 and mid-2024 across all search engines, aligning with the wider release of large-scale text-generation tools such as ChatGPT and GPT-4. DuckDuckGo showed the earliest and most visible rise in spam rates, while Bing experienced a delayed but sharper increase during early 2024. In contrast, Startpage maintained consistently lower spam levels, remaining below 2%, which can be attributed to its smaller crawl coverage and stricter indexing policies.

DuckDuckGo

The DuckDuckGo trend shows noticeable month-to-month variability in spam proportions, with short-term rises and dips rather than a consistent directional change over the observation period. In the initial phase, spam accounted for nearly half of all review results, stabilising around 45% as the total indexed reviews surpassed 4,500. From mid-2023, this proportion remained largely steady between 45–47%, maintaining a close balance with genuine reviews through mid-2024. This consistent pattern suggests an initial surge of automatically generated or low-quality content, followed by improved filtering and ranking adjustments that curbed spam visibility. The moderate decline after early 2024 points to DuckDuckGo's gradual adaptation to AI-driven spam detection and enhanced index refinement over time. The correlation analysis yielded a Pearson coefficient of $r = 0.049$ with $p = 0.838$, indicating an extremely weak positive association that is not statistically significant. This confirms that, despite small fluctuations, DuckDuckGo's overall spam ratio remained broadly stable over time with no meaningful linear trend.

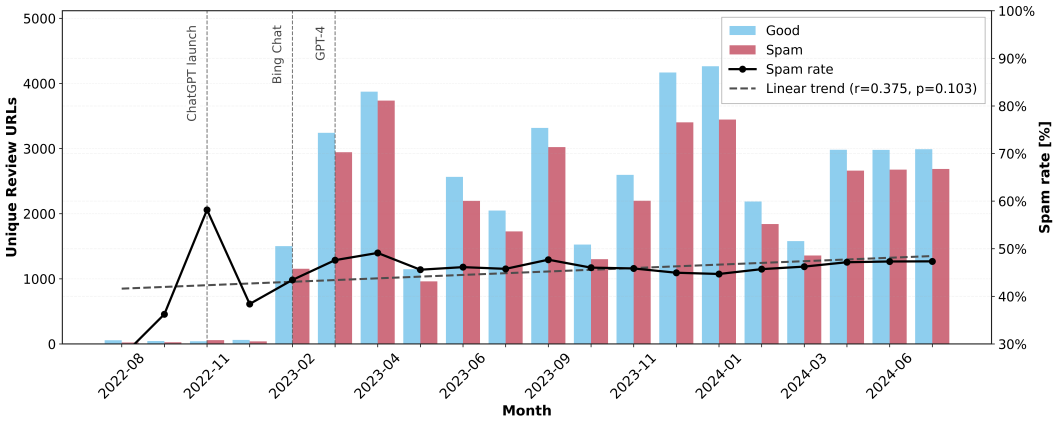


Figure 5.6: Temporal evolution of Spam vs. Good Reviews on Bing with Spam Rate (%)

Bing

The Bing dataset revealed a slight increase in spam-labelled reviews shortly after the introduction of large language models in early 2023. Spam prevalence rose from below 30% in mid-2022 to nearly 40% by February 2023, coinciding with the release of ChatGPT and Bing Chat integration. Between February and April 2023, both good and spam reviews grew rapidly, exceeding 3,500–3,700 unique review URLs per month. After mid-2023, the spam rate stabilised between 45–50%, maintaining near-equal proportions of genuine and spam content through mid-2024. This sustained high ratio indicates a prolonged presence of AI-assisted or promotional content in Bing's index, possibly driven by the large volume of user-generated material. The minor decline after mid-2024 suggests gradual moderation efforts or updated ranking algorithms that began reducing exposure to spam-heavy domains. The correlation analysis produced a Pearson coefficient of $r = 0.375$ with $p = 0.103$, indicating a very weak positive association that is not statistically significant. This means that although an early rise is visible which could be noise, there is no reliable linear trend across the whole period. The long-term spam ratio on Bing remained broadly stable, with fluctuations but no sustained upward or downward movement.

Startpage

Startpage displayed the most stable and consistent spam trends across the observation period. From mid-2022 onward, spam-labelled reviews generally outnumbered genuine reviews, fluctuating within a narrow range of 55–68%. Both review types appeared in similar volumes 2,000–3,000 URLs per month,

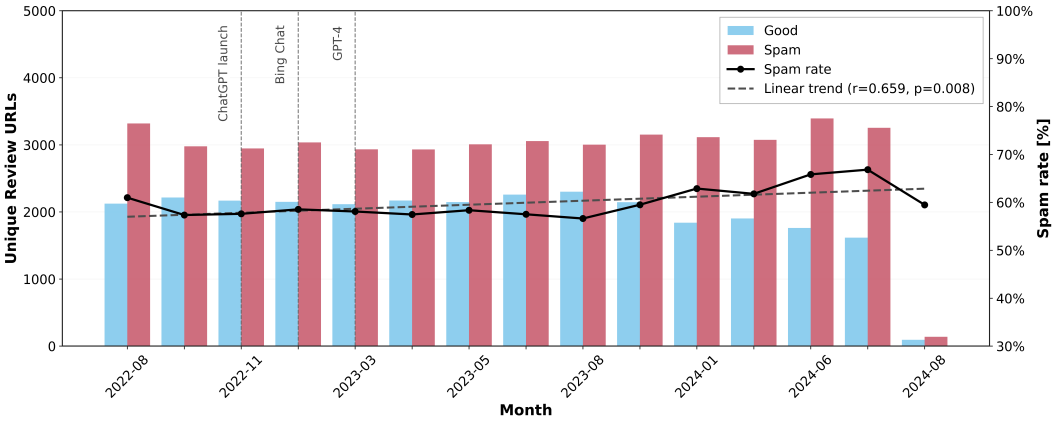


Figure 5.7: Temporal evolution of Spam vs. Good reviews on Startpage with corresponding Spam Rate (%).

with no abrupt spikes following major AI releases. The gradual rise observed in late 2023, peaking near 68% by mid-2024, indicates a steady but contained spread of templated or promotional content. Unlike Bing and DuckDuckGo, Startpage's relatively low volatility suggests smaller crawl coverage, stricter inclusion policies, and consistent moderation strategies that limited sudden influxes of spam text. The correlation analysis produced a Pearson coefficient of $r = 0.659$ with $p = 0.008$, signifying a moderate and statistically significant positive correlation between time and spam rate. This reflects a clear linear increase in spam proportion over the two-year period, confirming that Startpage gradually accumulated a higher share of spam content despite maintaining relatively stable total review volumes.

Result Interpretation and Discussion

The DistilBERT model demonstrated solid performance in both classification stages, achieving 78% accuracy in the binary stage and 63% in the multiclass stage. The slightly lower precision and F1-scores can be explained by several technical and data-related limitations.

First, the manually annotated dataset represented only a small portion of the overall corpus. This limited the model's exposure to diverse linguistic styles and borderline spam patterns, thereby constraining its ability to generalise to unseen examples. As a result, the learned decision boundaries were somewhat conservative and less adaptable to outlier cases.

Second, differences between human and model perception affected labeling consistency. Human annotators often relied on contextual or visual cues such

as webpage layout, advertisement banners, and formatting patterns to recognise spam, whereas the model analysed only plain text and linguistic features. Consequently, texts that appeared legitimate in wording but contained visual spam elements were correctly identified by humans but missed by the model.

Third, overlap between genuine and low-quality reviews introduced additional ambiguity. Many spam reviews were grammatically sound and used moderate sentiment, closely resembling authentic user feedback. This stylistic similarity explains the concentration of misclassifications near sentiment and tone boundaries observed in the confusion matrices.

Moreover, feature sparsity and template reuse across domains reduced the discriminative power of TF–IDF features. Although transformer embeddings captured deeper semantic relationships, they were less responsive to stylistic irregularities such as repetition, exaggerated emotion, or mechanical phrasing. This led to the model's prioritisation of recall for the spam class, adopting a cautious stance that limited false negatives at the cost of minor precision loss.

Overall, these outcomes illustrate the inherent challenge of identifying machine-generated spam content in large-scale, weakly supervised web data. Despite these constraints, the hybrid DistilBERT model provided a reliable and interpretable framework that scales effectively for longitudinal spam monitoring.

Chapter 6

Conclusion

This study analysed the transformation of online text from two perspectives: detecting AI-generated text and identifying spam or low-quality web reviews. Together, these analyses explored how automated text generation and optimisation practices that influence the reliability and authenticity of information on the web. The study combined interpretable linguistic features with transformer-based models to create scalable systems capable of assessing web content at large scale.

The first part focused on AI detection. Several models were tested, including Logistic Regression, Naïve Bayes, and Support Vector Machines using frequency-based features such as TF and TF-IDF. The linear Support Vector Machines with TF-IDF features achieved the best performance, reaching 99.54% accuracy on an 80-10-10 split. The findings showed that AI-generated text carries distinctive stylistic signatures such as frequent lexical repetition, uniform sentence rhythm, and limited grammatical variation, while human writing displays broader vocabulary and more varied structure. These consistent linguistic differences indicate that, under the conditions of this study, even fluent generative text remains statistically recognisable. Across all three search engines, Bing, DuckDuckGo, and Startpage, the temporal analysis shows a strong positive correlation between time and AI attribution (all $r > 0.90$, $p = 0$), indicating a consistent increase in AI-generated content from 2023 to 2024 despite overall rates remaining below 1%.

The second part addressed spam review detection using a manually verified dataset that combined user annotations, ChatGPT-based evaluations, and author verification. Out of 300 samples, 59 reached full agreement, 160 partial, and 81 disagreement. Human annotators showed the highest consistency, while AI alignment was stronger only in clear-cut cases. Several transformer models,

BERT-base, DistilBERT, and RoBERTa were evaluated, both as standalone encoders and as hybrid models combined with TF–IDF and linguistic features. The hybrid DistilBERT model achieved about 78% accuracy in review identification and 63% in spam classification. The performance shows that even strong transformer models struggle when spam closely resembles authentic user feedback. Spam content typically showed exaggerated sentiment, frequent adjectives, and very simple readability, while authentic reviews maintained a balanced tone and clearer structure. The outcomes reinforced the value of linguistic indicators such as part-of-speech ratios, readability, and sentiment scores for interpretable classification.

A longitudinal analysis from 2022 to 2024 showed an overall increase in spam content, with a noticeable rise in early 2023 followed by a period of stabilisation, possibly due to improved filtering by search engines. However, the temporal trend differed across search engines: Bing showed only a weak and non-significant increase ($r = 0.375$, $p = 0.103$), DuckDuckGo displayed almost no trend ($r = 0.049$, $p = 0.838$), while Startpage exhibited a moderate and statistically significant rise in spam rate over time ($r = 0.659$, $p = 0.008$). The measured correlation between AI-authored text and spam rate was very weak but statistically significant (Pearson's $r = 0.022$, $p < 0.001$), indicating only a minimal linear association. This suggests that, while both AI-generated content and spam increased over time, their relationship is limited and largely driven by independent factors rather than direct overlap.

Overall, the study shows that combining linguistic features with contextual embeddings provides a structured and scalable way to assess the authenticity and quality of web content. While the AI detection model performed strongly on the test set, its real-world generalisation remains uncertain without additional validation. In contrast, the review classification pipeline achieved reasonable performance in identifying review pages but showed limited accuracy when distinguishing between spam and good reviews, reflecting the inherent difficulty of separating well-written promotional spam from genuine user feedback.

For future work, it would be interesting to extend this system to multilingual datasets enriched with topic-based features. Developing adaptive models that learn continuously from user feedback for large-scale monitoring pipelines would help track emerging content behaviours and maintain transparency as generative technologies evolve.

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